

BINAY—Lighting Up The Future

Binay LED Aviation Obstruction Lights



**BINAY LED-based MEDIUM INTENSITY
Aviation Obstruction Light Beacon**

As per the International Civil Aviation Organisation (ICAO) requirements for use on structures above 45m

Available in Low Intensity and Medium Intensity versions (as per the International Civil Aviation Organization guidelines). BINAY's patented LED Aviation Lights come with a 5-year warranty

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave, and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime



**THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED
PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT**



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and can't be charged. One can do diagnostics, like with X-rays and MRI in hospitals, but for batteries instead.

Charge your devices with knee movements

Who needs chargers when the human body can charge the devices? Confused? Well, researchers at the Cranfield University and the University of Liverpool of Salford have devised an innovative way to generate power to fuel small gadgets. The device is known as the pizzicato knee-joint energy harvester. This is a circular device that fits onto the outside of the knee and consists of an outer ring fitted with 72 plectra and a central hub with four protruding arms.

Every time your knee bends, the outer ring rotates, causing the plectra to pluck the arms. This makes the arm vibrate like a guitar string, generating electrical energy. This electrical energy can then be used to charge devices.

According to the researchers, the device can currently harvest around two milliwatts of power but could exceed 30 milliwatts with a few improvements. This could enable a new generation of GPS tracking, more advanced signal processing, and more frequent and longer wireless transmission. The device could be particularly useful for soldiers who currently have to carry up to 10 kg of power equipment when on foot patrol.

T-shirt to charge your cellphone

The University of South Carolina's Xiaodong Li envisions a future where electronics is part of our wardrobe. He is working on the conversion of cotton T-shirt material into a smart device charger.

"We will soon see roll-up cellphones and laptop computers on the market. But a flexible energy storage device is needed to make this possible," said Li, a professor of mechanical engineering at the University.

Trying to be creative with a cotton T-shirt from a local discount store, Li's team soaked it in a solution of fluoride. After drying it, the team baked the T-shirt at high temperature. They excluded oxygen in the oven in order to prevent the material from charring or simply combusting. The surfaces of the resulting fibres in the fabric were shown by infrared spectroscopy to have been converted from cellulose into activated carbon. Yet, the material retained flexibility; interestingly, it could be folded without breaking.

The team then used nanoflowers of manganese oxide to cover the individual fibres in the activated carbon textile. This layer of manganese oxide was just a nanometre thick. Moreover, it enhanced the electrode performance of the fabric. This created a stable, high-performing supercapacitor.

By stacking these supercapacitors up, the researchers expect to be able to charge portable electronic devices such as cellphones.

Fuel cell that runs on glucose

MIT engineers have developed a glucose-based fuel cell for powering highly efficient brain implants of the future that